Econ 240B: Lecture 1

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This course will emphasize linear methods in econometrics. The main text for the first half is *Econometrics* by B.E. Hansen 2022.

Lecture 1 is mainly a recap of sample averages and different interpretations/methods of estimating the population mean of a distribution.

1 Sample Averages

First let's recall the definition of the sample average.

Definition 1. Sample average.

$$\bar{Y} = \frac{1}{N}\sum_{i=1}^{N} Y_i$$

where $Y_1, ..., Y_n$ are random variables is the sample average.

Consider the **Gaussian Location Model.** Suppose $Y_i \sim \text{iid}\mathcal{N}\left(\mu, \sigma^2\right)$ where both μ and σ^2 are unknown but what we really care about is μ. The hope is that Y_i in our sample is informative of μ.

Although we assume that σ^2 is unknown for now, we will change this later. The following discussion motivates why using the sample mean, $\bar{Y}$, is actually a good estimation for the population mean, μ.

2 Maximum Likelihood

In addition to the intuitive benefits of using the sample average as an estimator for the population mean, there is also good statistical reasoning for doing so. It turns out that the sample mean is also the **maximum likelihood (ML) estimator** of the population mean for the normal distribution. Within this framework, there are three ways to look at the problem.

2.1 Maximum Likelihood (ML)

As stated earlier, for the normal distribution,

$$\bar{Y} = \hat{\mu}_{ML}$$

the sample mean is the maximum likelihood estimator of μ. Given the two parameters of the normal, μ and σ^2, we know that the ML estimators are given by

$$\left(\hat{\mu}_{ML}, \hat{\sigma}^2_{ML}\right) = \arg\max_{\left(\mu_Y, \sigma^2_Y\right)} \ell\left(\mu_Y, \sigma^2_Y\right)$$

where $\ell\left(\cdot\right)$ is the log-likelihood function and is

$$\begin{aligned}\ell\left(\mu_Y, \sigma^2_Y\right) &= -\frac{n}{2}\ln\left(2\pi\sigma^2_Y\right) - \frac{1}{2\sigma^2_Y}\sum_{i=1}^{N}\left(Y_i - \mu_Y\right)^2 \\ &= \sum_{i=1}^{N}\left\{-\frac{1}{2}\ln\left(2\pi\sigma^2_Y\right) - \frac{1}{2\sigma^2_Y}\left(Y_i - \mu_Y\right)^2\right\}\end{aligned}$$

Since μ_Y and σ_Y^2 are both unknown, we solve for the system of equations such that the log-likelihood is maximized.

Oftentimes, the ML is the gold standard for any distribution. The caveat to using ML is that we need to know the distribution of the data with a certain number of known parameters.

However, we can try and ask how can we estimate information above and beyond what we know? This leads us to Oracle ML estimators.

2.2 Oracle ML

Oracle ML estimators tell us the estimator of a parameter *as if* we had information we actually don't.

Again, for the normal distribution we have that

$$\bar{Y} = \tilde{\mu}_{ML}$$

where

$$\tilde{\mu}_{ML} = \arg\max_{\mu_Y} \ell\left(\mu_Y, \sigma^2\right).$$

In this case, the problem is very similar but now we take the σ^2 as known. It turns out that regardless of the choice of σ^2, the only way to maximize the log-likelihood is by setting the estimator to the sample mean.

What if instead we want to treat σ_Y^2 is known, but don't actually know what it is?

2.3 "Quasi" ML

This perspective is in the spirit of "all models are wrong, but some are useful". Let consider what would happen if we arbitrarily guess the value σ^2 without caring about its accuracy. The question we seek to answer is how would this guess affect the estimator of the population mean?

It turns out that, again, for the normal distribution we will have

$$\bar{Y} = \bar{\mu}_{ML}$$

where

$$\bar{\mu}_{ML} = \arg\max_{\mu_Y} \ell\left(\mu_Y, \bar{\sigma}^2\right)$$

for some (possibly) incorrect σ^2.

In the case with the normal distribution, this will not prevent us from finding a reasonable estimator of μ. In fact, we still get the sample mean as the maximum likelihood estimator.

We can think of OLS as an interpretation of this type where we are assuming that the errors are normally distributed. When to use, or think about, estimators as standard ML, Oracle, or Quasi depends on the context.

2.4 Note on distributions

It's important to emphasize that we've been using the normal distribution so μ is orthogonal to the other parameters. This is the exception and not the rule. Therefore, from now on we will consider σ^2 as known.

3 Method of Moments

Maximum likelihood isn't the only way to retrieve estimators of population parameters. Probably the simplest way to think about estimators is to equation the sample moments to the theoretical moments of a distribution and solve for the parameters in question by using the **method of moments** (MM)

The method of moment estimator of the population mean for the normal distribution is again given by the sample mean,

$$\bar{Y} = \hat{\mu}_{MM}$$

where

$$\mu\left(\hat{\mu}_{MM}\right)=\bar{Y}$$

where $\mu\left(\cdot\right)$ is the **moment generating function**. In fact, the method of moment estimator of the population mean for any distribution will always be the sample mean.

When solving for the MM estimator, we will always have a system of K unknown parameters and K equations that correspond to the moment of each parameter we are interested in estimating.

4 General MM (GMM)

In this section, we considered a generalized version of the method of moments. In this setup, we can have more equations than unknowns and even when we have the same number of equations and unknowns. In this latter case, we introduce this notion of optimal weighting of moment conditions.

Again, for the normal distribution, we will have that

$$\bar{Y}=\hat{\mu}_{MM}$$

where

$$\frac{1}{N}\sum_{i=1}^{N}g\left(Y_i,\hat{\mu}_{MM}\right)=0$$

and

$$g\left(Y,\mu_Y\right)=Y-\mu_Y.$$

The goal is find $g\left(\cdot\right)$ to reduce the system to a model of observables and unknowns. The motivation for GMM is that it provides a framework foe handling endogenous regressors.

How can we tie together ML and GMM?

5 "Plug-in"/ "Analogy" Principle

For any cdf $F_y:\mathbb{R}\to[0,1]$, define the function

$$\mu\left(F_Y\right)=\int_{-\infty}^{\infty}ydF_Y\left(y\right)$$

Recall some basic properties of a cdf,

Fact 2. *If F_Y admits pdf f_y, then*

$$\mu\left(F_y\right)=...=\int_{-\infty}^{\infty}yf_Y\left(y\right)dy$$

if F_y admits pmf f_y, then

$$\mu\left(F_y\right)=\sum_{y\in\mathbb{R}}yf_y\left(y\right)$$

Let's note a couple observations that mark a key distinction between the field of econometrics and statistics.

5.1 Identification

Econometrics is concerned with the identification of parameters. We want to know whether we can express what we care about in terms of the distribution that we see in the data. In other ways, can the data help describe some theoretical underpinning of how I think the world works.

If F is the cdf of $Y_i \sim \mathcal{N}\left(\mu, \sigma^2\right)$, then

$$\mu\left(F\right) = ... = \mu$$

and what we care about is whether the parameter of interest, μ in this case, can be determined from the observed distribution, F.

5.2 Estimation

Since we do not observe F, we may want to consider the intuitive estimator of the cdf,

$$\hat{F}_N\left(y\right) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left(Y \leq y\right)$$

where $\hat{F}_N$ is the emperical cdf. Now the parameter of interest that we seek to estimate given the data can be described by

$$\mu\left(\hat{F}_N\right) = ... = \bar{Y}$$

5.3 Relating GMM and ML

5.3.1 GMM

Let's first consider the GMM in the continuous case,

Let $\mu\left(F_Y\right)$ satisfy

$$\int_{-\infty}^{\infty} g\left(y, \mu\left(F_Y\right)\right) dF_Y\left(y\right) = 0$$

and

$$g\left(y, \mu_Y\right) = y - \mu_Y$$

where again we define the functional as $\mu\left(\cdot\right)$ as above. Then

$$\mu\left(F\right) = ... = \mu$$

but it turns out the functional evaluated at the empirical cdf gives us our sample mean,

$$\mu\left(\hat{F}_N\right) = ... = \bar{Y}$$

5.3.2 Oracle ML

Now, let's consider the Oracle ML again where σ^2 is known. Define the function

$$\begin{aligned}\mu\left(F_Y\right) &= \arg\max_{\mu_Y} \int_{-\infty}^{\infty} \left\{-\frac{1}{2}\ln\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\left(y - \mu_Y\right)^2\right\} dF_Y\left(y\right) \\ &= \arg\min_{\mu_Y} \int_{-\infty}^{\infty} \left(y - \mu_Y\right)^2 dF_Y\left(y\right)\end{aligned}$$

We will end up getting

$$\mu\left(F\right) = ... = \mu$$

We can apply this functional to the empirical cdf and get that and

$$\mu\left(\hat{F}_N\right) = ... = \bar{Y}$$

5.3.3 Conclusion

This tells us that both the GMM and ML can be viewed as functionals of the CDF to define parameter estimates which also happen to coincide in the case with the normal distribution. Here we see that both GMM and ML can also be interpreted as "plug-in"/ "analogous" estimators.